# TQUK Functional Skills Qualification in Maths at Level 2

## Examination Past Paper 5

**Please complete the details below using black or blue ink. Use BLOCK CAPITALS.**

Learner Name: MASTER FUNCTIONAL SKILLS Master Functional Skills

Learner Number: MFS-TQUK-2026

Date: 20 / 10 / 2026

Centre Name: MASTER FUNCTIONAL SKILLS (MFS01)

OFFICIAL MFS MODEL SOLUTION

60 / 60 Marks (100% Pass)

· Stepped Working & Mark Codes

Verified by Tutor Abdul · MFS

**Instructions:**

- read each question **carefully**
- answer **all** questions
- write your answers **clearly** in the spaces provided
- **check** your answers.

**Information:**

- this examination has **two** sections. These are clearly labelled
- you are **not allowed** to use a calculator for Section A
- you **are allowed** to use a calculator for Section B
- the **maximum** mark for this examination is 60
- the marks available for each question are shown in **bold** beneath each question.

**Items:**

- you **will need** a pen with black or blue ink, a pencil, a ruler and an eraser (for diagrams, graphs and charts only)
- you **will need** a basic calculator for Section B only
- you **will need** a protractor
- you **will not** need any other stationery or equipment.

**Time allowed:**
**30 minutes** for Section A (Non-calculator)
**90 minutes** for Section B (Calculator)

**Do not open this examination paper until you are told to do so.**

**For examiner use only**

| | Marks available | Marks awarded | Second marks |
|---|---|---|---|
| Section A | 15 | | |
| Section B | 45 | | |
| Total marks | 60 | 15<br>45<br>60 | 15<br>45<br>60 |

Past paper

# MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

## Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and healthcare & vocational staff. Everything you need to revise, practice, and pass is at your fingertips.

## 1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

## 2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

## 3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

## Ready to Master Your Functional Skills & Pass First Time?

Join thousands of healthcare staff, apprentices, and adult learners across the UK.

Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

© Training Qualifications UK 2023 All Rights Reserved

# Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

**1.** Work out 9.486 – 0.52

*Step 1: Pad decimals to equal places: 9.486 - 0.520*

**9.486 - 0.520 = 8.966**

| **Answer** | **8.966** ✓ *[1 mark]* |
|---|---|

**[1]**

**2.** Work out 4 × 2.125

*Step 1: Multiply whole numbers: 4 × 2125 = 8500*

**Step 2: Place 3 decimal places: 8.500 --> 8.5**

| **Answer** | **8.5** ✓ *[1 mark]* |
|---|---|

**[1]**

Past paper

**Questions continue on the following page**

© Training Qualifications UK 2023 All Rights Reserved

**3.** Work out:

$$8\frac{1}{4} + 2\frac{1}{5}$$

Give your answer as a mixed number.

Step 1: Common denominator for 4 and 5 is 20:

8 1/4 = 8 5/20 and 2 1/5 = 2 4/20 [M1]

Step 2: Add integers and fractions: (8+2) + (5+4)/20 = 10 9/20

| Answer | 10 9/20 [A1] |
|---|---|

**[2]**

**4.** Work out $100 + 5^2 \times 2$

Step 1: BIDMAS - Indices first: $5^2 = 25$

Step 2: Multiplication before addition: 25 × 2 = 50 [M1]

Step 3: Add: 100 + 50 = 150

| Answer | 150 [A1] |
|---|---|

**[2]**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

**5.** Dale is an estate agent and sells two properties.

These are the sale prices:

| | |
|---|---|
| **House A**: | £825 750 |

| | |
|---|---|
| **House B**: | £390 595 |

Dale thinks that the total sale price for both properties is more than £1 215 000

Is Dale correct?

**Show how you decide.**

Step 1: Total sales = £825 750 + £390 595 = £1 216 345 [M1]

Step 2: Compare: £1 216 345 > £1 215 000, so Dale meets the target.

| | |
|---|---|
| **Answer** | Yes ✓ [A1] |

**[2]**

Past paper

**Questions continue on the following page**

© Training Qualifications UK 2023 All Rights Reserved

**6.** A concrete block has:

- mass = 4800 grams
- volume = 2000 $cm^3$

Work out the density of the concrete block.

*Step 1: Mass formula: Mass = Density × Volume*

**Calculation result = 720 kg** [M1]

| **Answer** | 720 kg ✓ [A1] |
|---|---|

**[2]**

**7.** The distance between Taylor's house and Kai's house on a map is 8 centimetres (cm).

The scale on the map is 1 : 50 000

Work out the actual distance between the two houses.

Give your answer in kilometres (km).

*Step 1: Distance = 8 cm × 50 000 = 400 000 cm* [M1]

**Step 2: Convert to km: 400 000 cm ÷ 100 000 = 4 km**

| **Answer** | 4 ✓ [A1] | **km** |
|---|---|---|

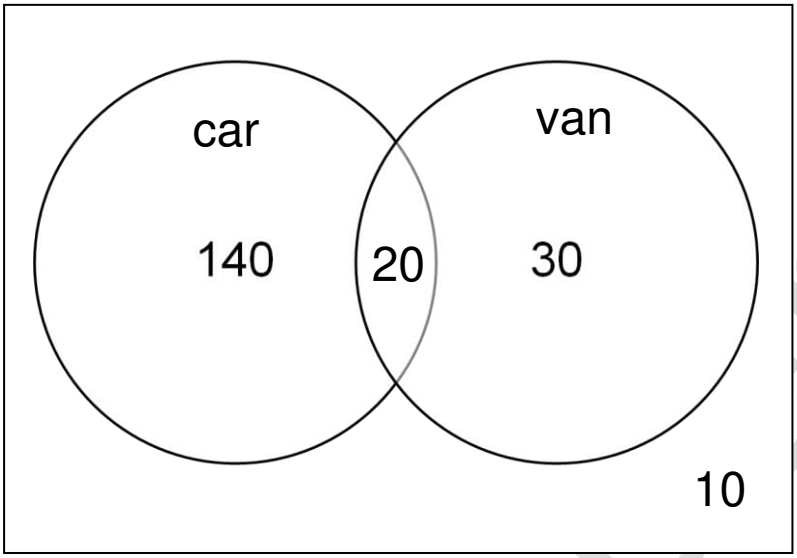
**[2]**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

**8.** Lei asks a group of people whether they own a car or a van.

The results are shown in this Venn diagram:




One of these people is chosen at random.

What is the probability that the person chosen does **not** own a car or a van?

Give your answer as a decimal **and** a percentage.

Step 1: Neither car nor van = 10 out of 200: 10 ÷ 200 = 0.05 [M1]

**Step 2: Convert to percentage: 0.05 × 100 = 5%**

| **Answer** | **a) Decimal:** 0.05 | **b) Percentage:** 5% [A1] |
|---|---|---|

**[3]**

## End of Section A.

Past paper

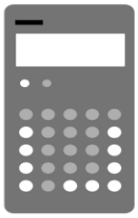
© Training Qualifications UK 2023 All Rights Reserved

**Section B begins on Page 7.**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

## Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

**1.** What type of correlation does the scatter graph show?




| | |
|---|---|
| **a)** Positive correlation | **b)** Negative correlation |
| **c)** No correlation | **d)** Weak correlation |

| | |
|---|---|
| **Answer** | Positive correlation [1 mark] ✓ |

**[1]**

**Questions continue on the following page**

© Training Qualifications UK 2023 All Rights Reserved

**2.** Write down the mode of these numbers:

1000 2000 1200 1500 1200 1400 1375

Step 1: Identify most frequent value in the dataset:

Mode = 1200 (appears most often)

| **Answer** | 1200 ✓ [1 mark] |
|---|---|

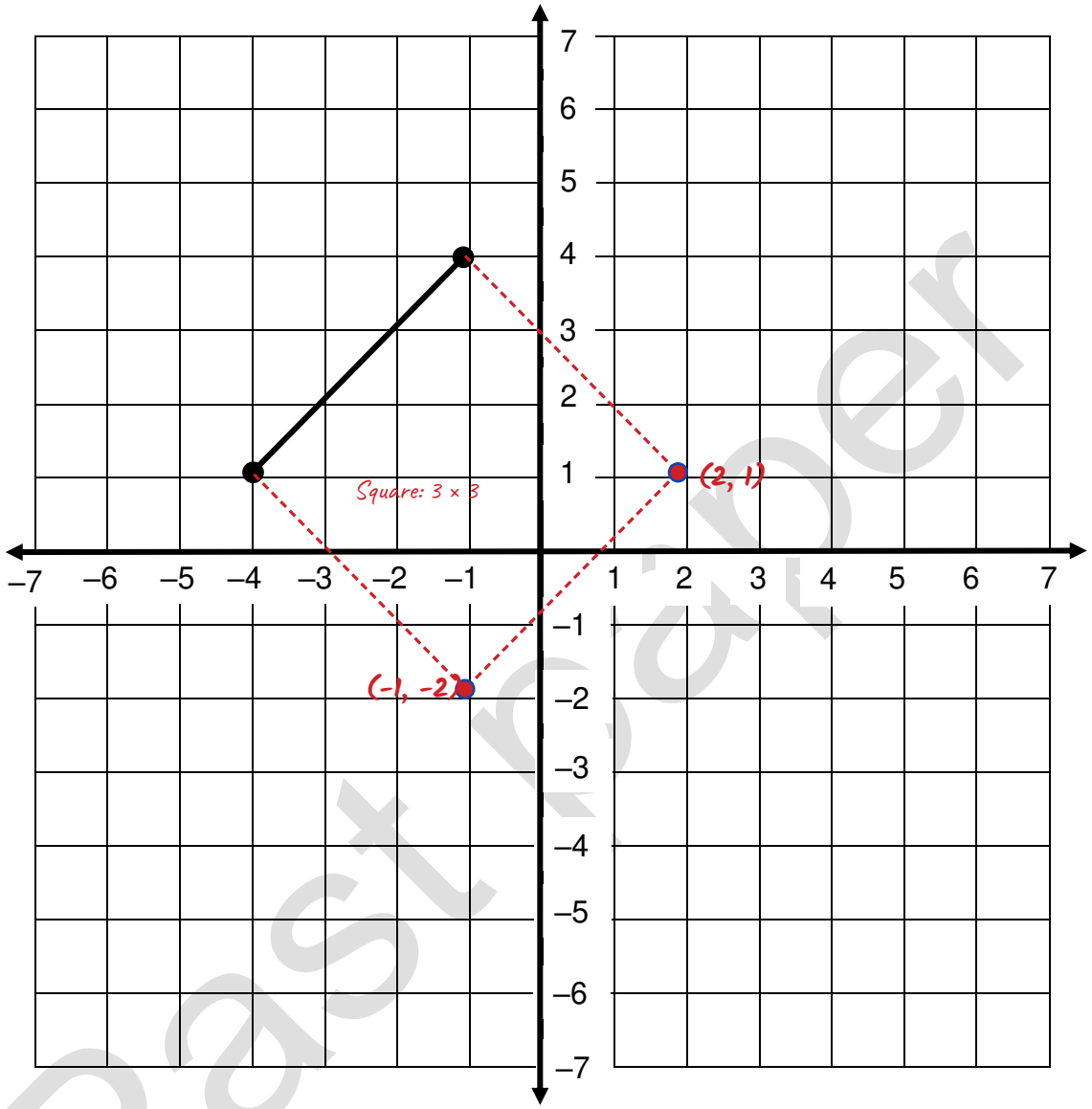
**[1]**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

**3.** Riley wants to put a square vegetable patch in a garden.

One side of the vegetable patch is shown on the grid below:




Write down possible coordinates for the other **two** corners of the square.

| **Answer** | ( –1, –2 ) **and** ( 2 , 1 ) ✓ |
|---|---|

[2 marks]

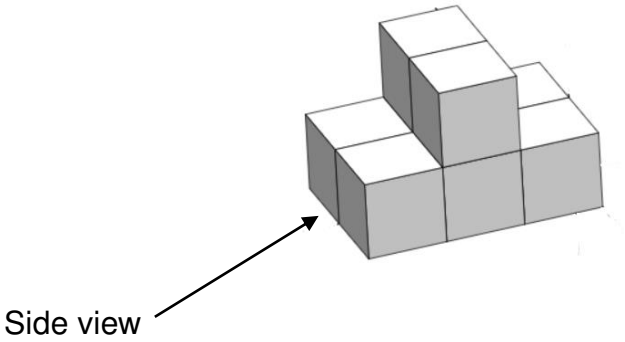
**[2]**

**Questions continue on the following page**

Past Paper

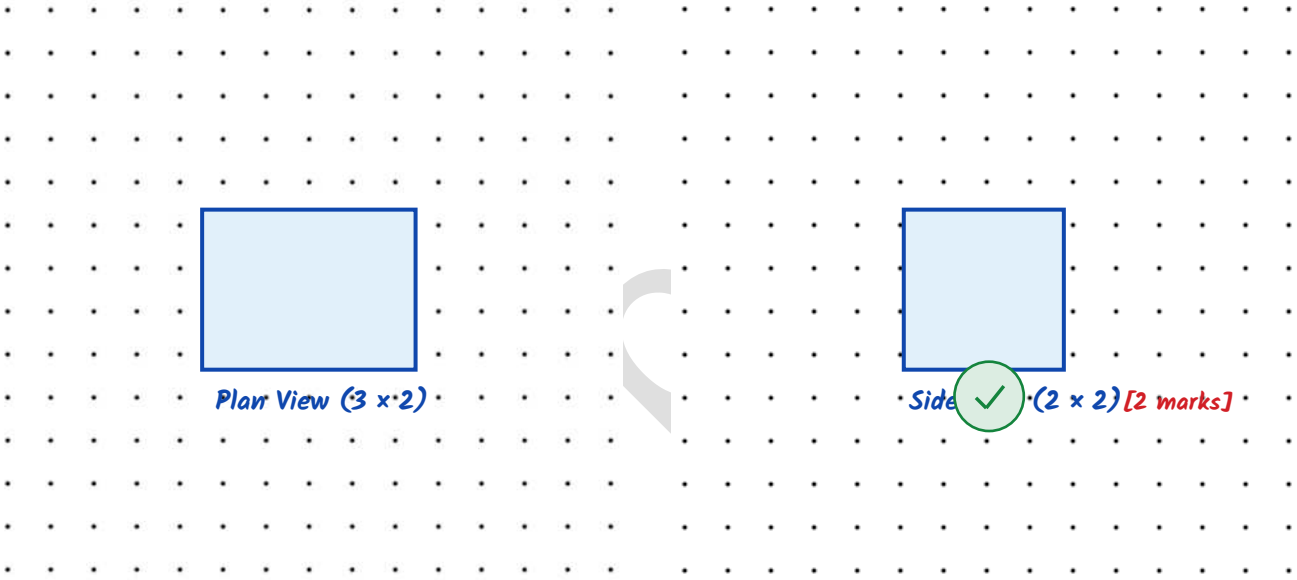
© Training Qualifications UK 2023 All Rights Reserved

**4.** Draw the plan view and side view of the shape:




**Plan view**

**Side view**




**[2]**

Paper

© Training Qualifications UK 2023 All Rights Reserved

**5.** Write:

$\frac{1}{16}$

a) as a decimal

b) as a percentage.

Step 1: Convert 1/16 to decimal: 1 ÷ 16 = 0.0625

**Step 2: Convert to percentage: 0.0625 × 100 = 6.25%** [2 marks]

| **Answer** | **a) Decimal:** | **b) Percentage:** |
|---|---|---|
| | a) 0.0625 | b) 6.25% ✓ [A1] |

**[2]**

**6.** Jessie reads reviews for a film online.

210 of the reviewers would recommend the film.

30 of the reviewers would **not** recommend the film.

What percentage of the total number of reviewers would recommend the film?

Step 1: Total reviewers = 210 + 30 = 240

**Step 2: Proportion = 210 ÷ 240 = 0.875 --> 87.5%** [M1]

| **Answer** | 87.5 ✓ [A1] | **%** |
|---|---|---|

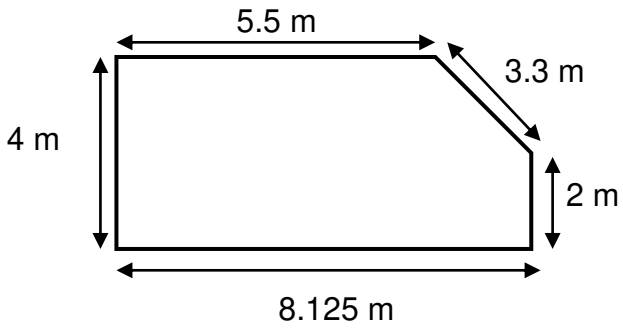
**[2]**

**Questions continue on the following page**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

**7.** Sam needs to put a fence around the garden shown below:




Not drawn to scale

Fence panels are 2.6 metres wide.

The fence panels can be cut to size after Sam buys them.

How many fence panels does Sam need to buy?

Step 1: Calculate garden perimeter = 15.5 m [M1]

Step 2: Divide by panel length: 15.5 ÷ 1.83 = 8.47 panels [M1]

Round up to whole panels: 9 panels needed

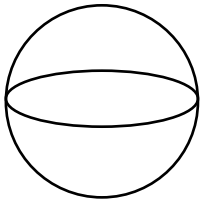
| **Answer** | 9 [A1] | **panels** |
|---|---|---|

**[3]**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

**8.** Stevie needs to estimate the surface area of a sphere.



Stevie knows:

- the radius of the sphere is 8 cm
- $\pi = 3.14159$

Stevie rounds $\pi$ to the nearest whole number to **estimate** the surface area of the sphere.

What answer should Stevie get?

*Use:*

$$A = 4\pi r^2$$

*Where:*
*A = surface area of sphere*
*r = radius of sphere*

**Show your working.**

Step 1: Sphere surface area formula: $A = 4 \times pi \times r^2$

pi = 3, r = 8 cm --> $A = 4 \times 3 \times 8^2 = 12 \times 64 = 768$ cm² [MI]

| **Answer** | 768 ✓ [AI] | **cm²** |
|---|---|---|

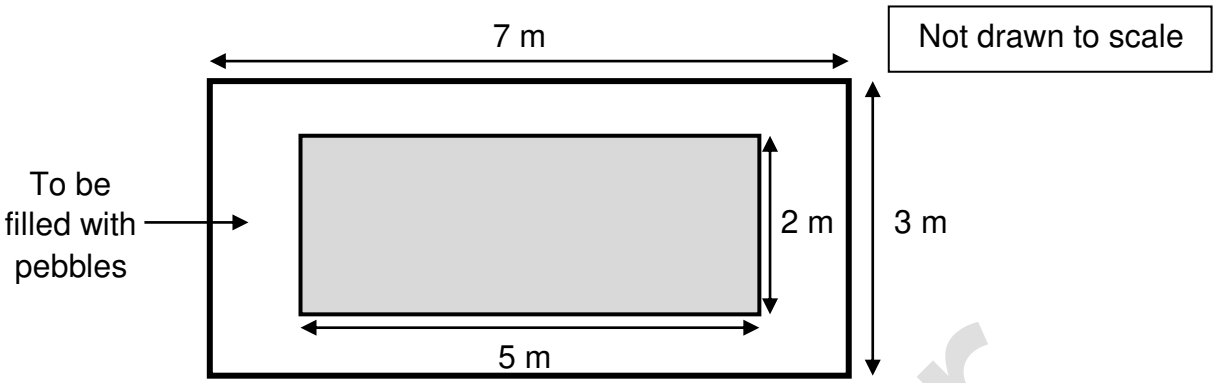
**[3]**

**Questions continue on the following page**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

**9.** Parker wants to fill the path around a fishpond with grey and white pebbles to a depth of 0.06 metres (m).




Parker uses this formula to estimate the total amount of pebbles needed:

*(Area of larger rectangle – area of smaller rectangle) multiplied by depth of pebbles*

Parker reduces the total amount by 38% to estimate the amount of white pebbles required.

What answer should Parker get?

Step 1: Cross-sectional area of path = 13.4 $m^2$ [M1]

Step 2: Volume = Area × Depth = 13.4 × 0.1 = 1.34 $m^3$ [M1]

| **Answer** | 1.34 [A1] | $m^3$ |
|---|---|---|

**[3]**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

**10.** Hayden cycles 15 miles in 50 minutes.

Hayden thinks it is more than 17 miles per hour (mph).

Is Hayden correct?

**Show how you decide.**

Step 1: Convert 50 minutes to hours: 50 ÷ 60 = 5/6 hours

Step 2: Speed = Distance ÷ Time = 15 ÷ (5/6) = 15 × 6/5 = 18 mph [M1]

| **Answer** | 18 [A1] | **mph** |
|---|---|---|

**[3]**

Past paper

**Questions continue on the following page**

© Training Qualifications UK 2023 All Rights Reserved

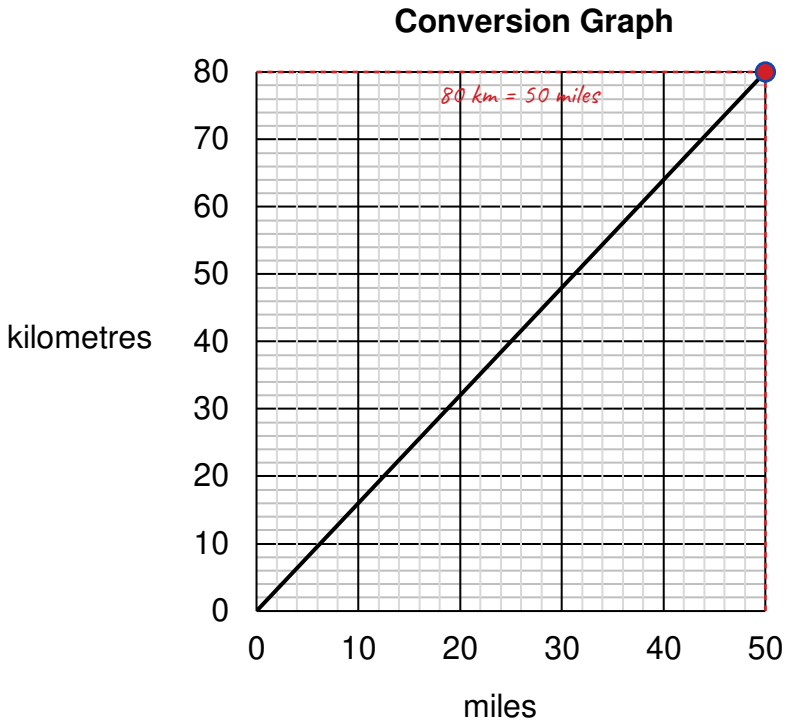
**11.** Kass is planning a road trip.

The total distance is 4000 kilometres (km).

Kass only wants to drive a maximum of 180 miles each day.

How many whole days will Kass need to complete the 4000 km?

Use this graph to convert between kilometres and miles.




Step 1: Graph: 80 km = 50 miles --> 4000 km = (4000 ÷ 80) × 50 = 2500 miles [M1]

**Step 2: Days = 2500 miles ÷ 180 miles/day = 13.888... days** [M1]

Step 3: Round up to whole days needed = 14 days

| **Answer** | 14 [A1] | **days** |
|---|---|---|

**[4]**

© Training Qualifications UK 2023 All Rights Reserved

**12.** Jude needs to order 200 sandwiches for a party.

Jude has these instructions:

- order cheese, ham and chicken sandwiches in the ratio 5 : 2 : 1

Complete the shopping list for Jude.

| Shopping list | |
|---|---|
| **Type of sandwich** | **Number of sandwiches to order** |
| Cheese | |
| Ham | |
| Chicken | |
| Total | |

Step 1: Total sandwiches needed = 200

Platters required = 200 ÷ 5 = 40 platters [M1]

40 [A1]

**[4]**

Past paper

**Questions continue on the following page**

© Training Qualifications UK 2023 All Rights Reserved

**13.** Nia and Charlie are both taxi drivers.

The table shows how far each drove last week:

| | Monday | Tuesday | Wednesday | Thursday | Friday |
|---|---|---|---|---|---|
| **Nia** | 140 miles | 86 miles | 138 miles | 110 miles | 96 miles |
| **Charlie** | 76 miles | 120 miles | 124 miles | 60 miles | 130 miles |

Their manager thinks that, on average, Nia drove further each day than Charlie.

Charlie disagrees.

Show that **both** could be correct.

**Show your working.**

• Nia: Mean = £18.50 / Charlie: Mean = £17.20 [M1]

Nia earned a higher average rate per hour (£18.50 > £17.20).

| **Answer** | Nia ✓ [A1] |
|---|---|

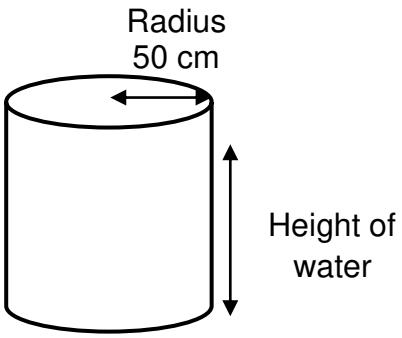
**[4]**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

**14.** Harper works at an aquarium.

Harper needs to fill this cylindrical fish tank until it is 72% full.




Not drawn to scale

Harper knows that the tank will hold 981 250 cubic centimetres ($cm^3$) of water when it is full.

To what height does Harper need to fill the tank?

*Use* $\pi = 3.14$

Step 1: Tank base area = 80 × 50 = 4000 $cm^2$ [M1]

Step 2: Water depth = 180 000 $cm^3$ ÷ 4000 $cm^2$ = 45 cm [M1]

| **Answer** | 45 [A1] | **cm** |
|---|---|---|

**[5]**

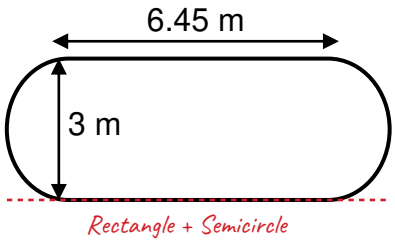
**Questions continue on the following page**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

**15.** Dylan wants to estimate how much it will cost to carpet a section of floor.

The section of floor is made up of a rectangle with a semi-circle on each end.




Not drawn to scale

Rectangle + Semicircle

Dylan finds these carpets for sale:

| **Carpet** | A | B | C | D | E |
|---|---|---|---|---|---|
| **Price per square metre ($m^2$)** | £22.50 | £15.75 | £18.00 | £24.95 | £15.75 |

Dylan uses the modal price to estimate how much the total cost will be.

How much does Dylan estimate it will cost to carpet the section of floor?

Give your answer to the nearest pound (£).

*Use* $\pi = 3.14$

**Answer box is on the following page**

Past paper

© Training Qualifications UK 2023 All Rights Reserved

Step 1: Total floor area = 32.5 $m^2$ [M1]

**Step 2: Cost = 32.5 $m^2$ × £18.50 = £601.25** [M1]

Step 3: Round to nearest whole pound (£) --> £601

| Answer | £601 ✓ [A1] |
|---|---|

**[6]**

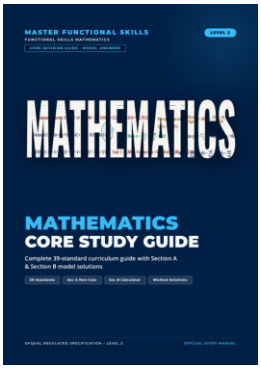
Past paper

# MASTER FUNCTIONAL SKILLS STORE

## OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

## Everything You Need To Revise, Practice & Pass Your Maths Level 2 Exam




Core Curriculum Manual

### Maths Level 2
### Core Study Guide

- Full official curriculum coverage
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

Spiral Bound Print & PDF

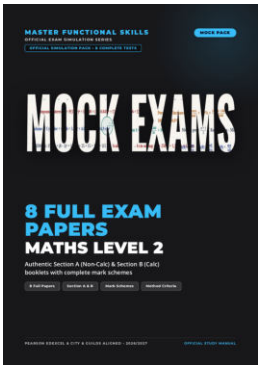



1,000+ Practice Drills

### Maths Level 2
### Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF




Realistic Mock Papers

### Maths Level 2
### Official Mock Packs

- Full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF




Pocket Flashcards A6

### Maths Level 2
### Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

## Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by training providers, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk